

Real gases

Learning outcomes

By the end of this section you should be able to explain the limitations of the ideal gas model.

We have seen in the previous section that ideal gases are assumed to consist of particles that have negligible volume and negligible attractive forces. Under certain conditions, this can be quite close to reality. However, is this true under all conditions? Can the assumptions still hold when the gas is cooled down or when the gas is pressurised?

All gases deviate from ideal behaviour to some extent and are described as real gases. With real gases, some of the assumptions made about ideal gases no longer apply under certain conditions.

Nature of Science

Aspect: Patterns and trends

The gas laws apply only to ideal gases. How appropriate is it to apply such laws to real gases? Under what conditions do the variations between real and ideal gases become significant?

At low pressure, gas particles are far apart and the forces of attraction between them are negligible, so the gas acts as an ideal gas. At very high pressure the gas particles are closer together and are influenced by forces of attraction. As such, the assumption made in the ideal gas model no longer holds, and the gas cannot be described as an ideal gas anymore. At high pressures, the gas is no longer behaving as an ideal gas and is called a real gas.

At low temperatures, the particles move less rapidly (have lower average kinetic energy). This means that there is a greater opportunity for intermolecular forces between the particles to have an effect. As such, the intermolecular forces of attraction cannot be assumed to be negligible, and the assumption made in the

ideal gas model no longer holds. The gas cannot be described as an ideal gas anymore. At low temperatures, the gas is no longer behaving as an ideal gas and is called a real gas.

In fact, high pressures and low temperatures are precisely the conditions under which gases can be liquified. The ideal gas model can no longer be used to describe the behaviour of the particles in a gas.

Study skills

Real gases deviate from ideal gas behaviour under two conditions: at high pressures and low temperatures.

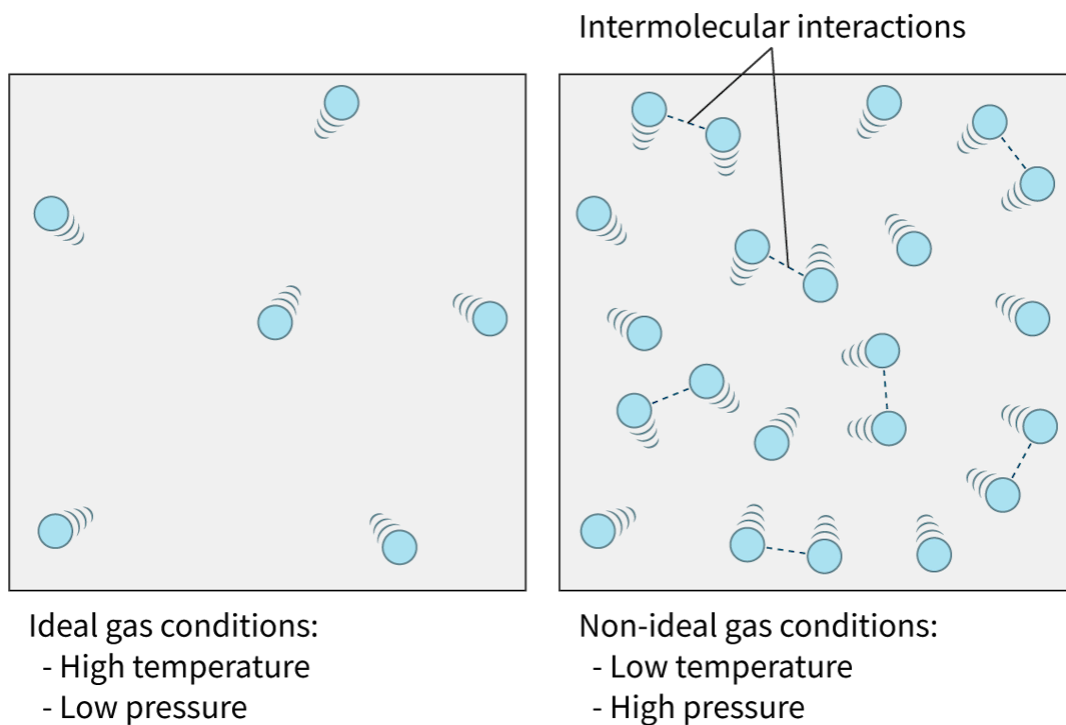


Figure 1. A comparison of the behaviour of a real and an ideal gas.

 More information for figure 1

The image displays two side-by-side diagrams comparing the behavior of an ideal gas and a real gas. On the left, the ideal gas is represented by uniformly distributed particles, illustrated as blue circles with evenly spaced motion lines indicating random movement and no interactions between them. On the right, the real gas is depicted with more densely packed particles, also shown as blue circles, but with dashed lines between them indicating interactions or collisions. The real gas particles show movement lines, suggesting more complex, less uniform motion compared to the ideal gas.

[Generated by AI]

Would all gases deviate from ideal gas behaviour in the same way?

Making connections

Section S2.2.8 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/intermolecular-forces-id-44359/>). — Under comparable conditions, why do some gases deviate more from ideal behaviour than others?

Since all substances contain different atoms and bonds, they also vary in their intermolecular forces of attraction. Similar substances have the same type of intermolecular forces of attraction. This will be covered more in depth later in the course, but for now, you need to know that for similar substances having the same type of intermolecular forces of attraction, the strength of these attractions will increase as molar mass increases. Which gases will behave more like ideal gases? The ones with smaller or larger molar mass? Think about people in a crowded room: which persons would be able to move through with greatest speed? The smallest people will be able to move through with greatest speed, avoiding collisions with other people. This is true for gas particles as well: the ones that are 'smaller' or lighter are the ones that will move faster. Therefore, gases that show the most ideal behaviour are those with a low molar mass and the weakest intermolecular forces between the particles. For example, among the noble gases, helium has the lowest molar mass and the weakest intermolecular forces of attraction. As such, its behaviour will be closest to the ideal gas behaviour from among the noble gases, since in an ideal gas, the intermolecular forces of attraction are negligible.

Theory of Knowledge

Helium behaves as an ideal gas under a range of temperature and pressure conditions due to its weak intermolecular forces. In addition to filling balloons, helium is used in many industrial applications including medical imaging, semiconductor chips in computers and air bags. Although highly abundant in space, helium is produced by only eight countries in the world. Helium is typically captured as a by-product during the processing of natural gas and once it enters the air, it is nearly impossible to recapture — because it behaves so well as an ideal gas. The global abundance of helium has been declining since 2006, causing massive shortages resulting in delays in manufacturing and medical

treatment worldwide. To what extent should governments be held responsible for the worldwide production and consumption of helium in the face of a shortage?

Activity

- **IB learner profile attribute:**
 - Open-minded
 - Inquirers
- **Approaches to learning:**
 - Thinking skills
 - Being curious about the natural world
 - Asking questions
 - Designing procedures and models
 - Applying key ideas and facts in new contexts
 - Engaging with and designing linking questions
 - Social skills — Actively seeking and considering the perspective of others
- **Time required to complete activity:** 30 minutes
- **Activity type:** Pairs activity

In pairs, you will be making a Brownian motion generator box to observe the effect of increasing pressure, volume and temperature on the behaviour of the gas particles.

1. Find a small cardboard box — of half the size of a shoe box.
2. Place some marbles in the box.
3. Shake the box. Observe how the marbles behave. Do you think this model describes the behaviour of particles in an ideal gas? Discuss with your partner.
4. Increase the pressure by decreasing volume by adding a divider or by adding more marbles. How does the increase in pressure affect the gas particles? Do they behave like an ideal gas more or less with the increase in pressure? Discuss with your partner.
5. Temperature can be decreased by shaking the box less, since temperature affects the kinetic energy of the particles. How does the decrease in temperature affect the gas particles? Do they behave like an ideal gas more or less with the decrease in temperature? Discuss with your partner.

You should be able to see how the particles will then be closer to one another, therefore having less space between them and allowing for more interactions. This creates a larger deviation from ideal gas behaviour.